

EMAIL AUTOMATION PROJECT

1. Executive Summary

Project Name

Email Automation System

Objective

To automate the process of generating, personalizing, scheduling, and sending emails to customers, employees, vendors, or stakeholders while minimizing manual effort, reducing errors, and improving communication efficiency.

The solution aims to:

- Eliminate repetitive manual emailing tasks
- Improve operational efficiency
- Ensure consistent communication
- Reduce turnaround time
- Enable scalable outreach without increasing manpower

2. What is Email Automation?

Email Automation is the process of automatically sending emails based on predefined business rules, schedules, events, or data triggers without requiring manual intervention.

Instead of manually drafting and sending hundreds of emails, the system automatically:

- Reads recipient information
- Personalizes email content
- Sends emails at the required time
- Tracks delivery status
- Maintains communication logs

Example :

Without Automation:

HR executive manually sends offer letters to 50 candidates.

With Automation:

The system reads candidate data from a database/spreadsheet and automatically sends personalized offer letters to all candidates.

3. Why is Email Automation Necessary?

Current Challenges

Organizations often face:

Manual Effort

Employees spend significant time sending repetitive emails.

Human Errors

Common mistakes include:

- Missing recipients
- Wrong attachments
- Incorrect names
- Duplicate emails

Lack of Scalability

As customer or employee volume increases, manual communication becomes difficult.

Delayed Communication

Important updates may not reach recipients on time.

Poor Tracking

Manual processes make it difficult to know:

- Who received emails
- When emails were sent
- Whether communication was completed

4. Business Problem Statement

Existing Situation

Business teams frequently send emails for:

- Customer onboarding
- Follow-ups
- Reports
- Notifications
- Reminders
- Marketing campaigns
- Internal communications

These processes are currently manual and consume valuable operational resources.

Business Impact

- Increased operational cost
- Reduced employee productivity
- Communication delays
- Inconsistent customer experience
- Risk of human errors

Proposed Solution

Implement an automated email system that generates and sends emails based on business rules and centralized data sources.

5. Technology Stack Evaluation and Solution Recommendation

Option 1: Google Sheets + Google Apps Script (Recommended)

Technology Stack

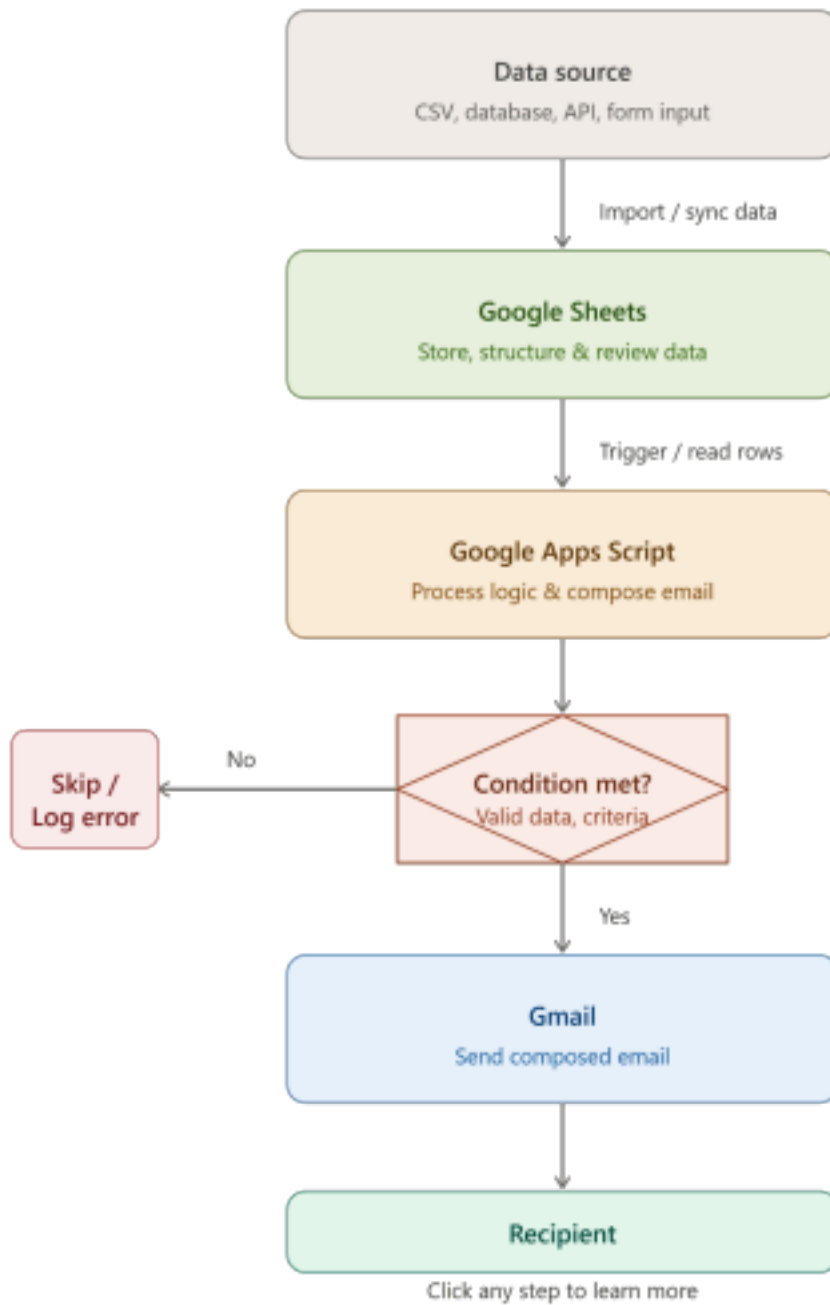
- Google Sheets
- Google Apps Script
- Gmail
- Google Workspace

Advantages

- Lowest implementation cost
- Rapid development
- No server infrastructure required
- Easy maintenance
- Business users can update recipient data directly
- Native integration with Gmail

Architecture Flow

Data Source → Google Sheets → Apps Script → Gmail → Recipient



Limitations

- Gmail sending limits
- Suitable for small to medium email volumes
- Limited advanced analytics

Best Fit

Organizations needing quick automation with minimal investment.

Option 2: Custom Web Application

Technology Stack

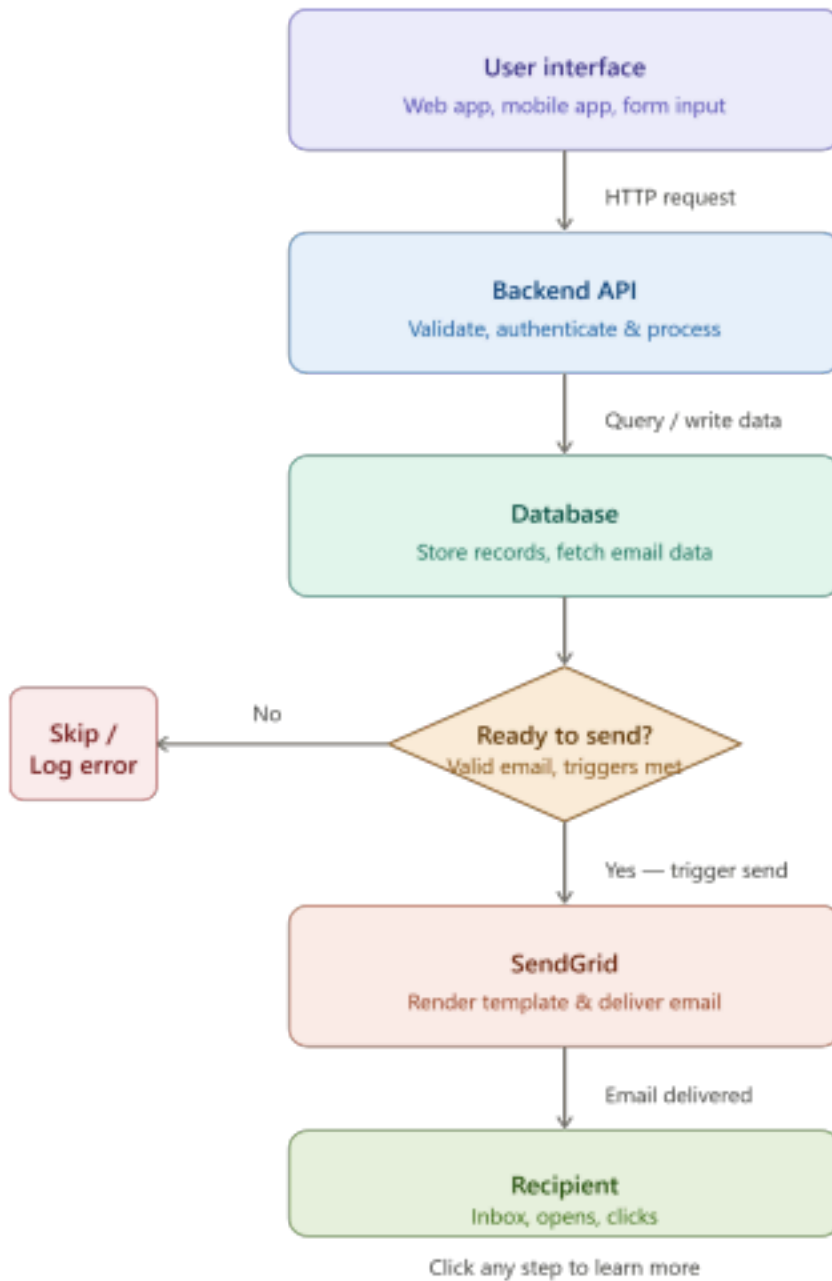
- React (Frontend)
- Node.js (Backend)
- PostgreSQL (Database)
- SendGrid (Email Service)

Advantages

- Fully customizable
- Better scalability
- Centralized management
- Advanced reporting and dashboards
- Better role-based access control

Architecture Flow

User Interface → Backend API → Database → SendGrid → Recipient



Limitations

- Higher development cost
- Longer implementation timeline
- Requires server maintenance

Best Fit

Organizations expecting significant growth and requiring advanced automation capabilities.

Option 3: Cloud-Native Enterprise Solution

Technology Stack

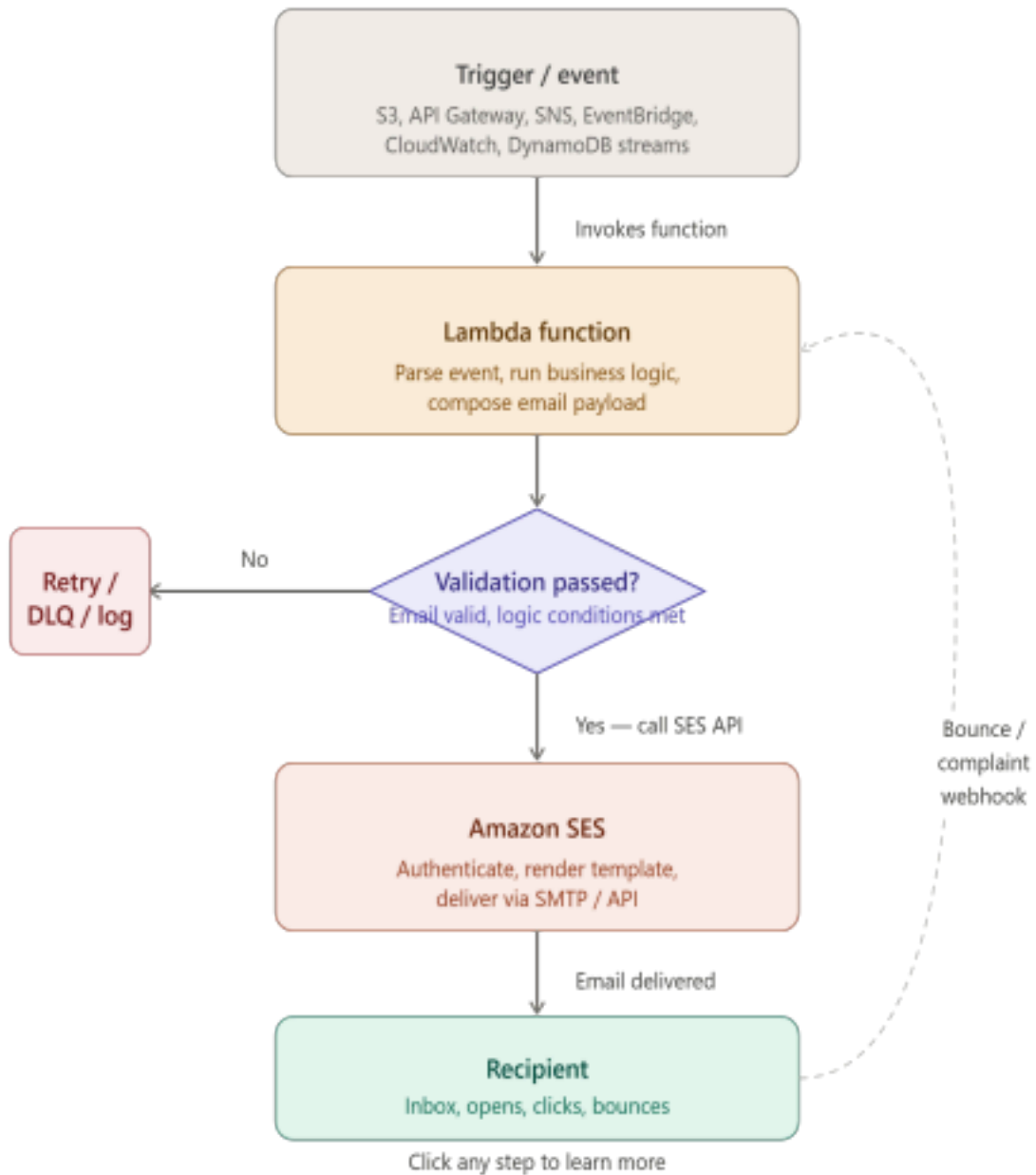
- AWS Lambda
- Amazon SES
- DynamoDB
- CloudWatch

Advantages

- Highly scalable
- Enterprise-grade reliability
- Cost-efficient at high email volumes
- Event-driven architecture

Architecture Flow

Trigger/Event → Lambda Function → SES → Recipient



Limitations

- Higher technical complexity
- Requires cloud expertise
- Longer setup time

Best Fit

Large organizations with high email volumes and enterprise-level automation requirements.

Use this revised comparison table:

Criteria	Google Sheets + Apps Script	Custom Application	AWS Cloud Solution
Implementation Cost	Low	Medium	High
Maintenance Effort	Low	Medium	Medium
Scalability	Medium	High	Very High
Infrastructure Requirement	None	Required	Required

Customization Capability	Medium	High	Very High
Technical Complexity	Low	Medium	High
Operational Overhead	Low	Medium	High
Business User Accessibility	High	Medium	Low
Enterprise Readiness	Medium	High	Very High

Final Recommendation

Recommended Solution: Google Sheets + Google Apps Script

Justification

After evaluating multiple implementation approaches, the Google Workspace-based solution is recommended because:

- It effectively addresses the current business requirement of automating repetitive email communication.
- It requires minimal infrastructure investment.
- It offers a cost-effective implementation model.
- It reduces operational overhead and maintenance effort.
- It provides sufficient scalability for current business needs.
- It supports future migration to more advanced architectures if business requirements expand.

6. Conclusion

- The Email Automation Project aims to streamline and automate repetitive email communication processes within the organization. By replacing manual email handling with an automated workflow, the solution will improve operational efficiency, reduce human errors, ensure timely communication, and enhance overall productivity.
- After evaluating multiple technology approaches, including a Custom Web Application and a Cloud-Native Enterprise Solution, the Google Sheets and Google Apps Script approach has been identified as the most suitable option for the current business requirements. This solution offers a cost-effective, easy-to-maintain, and scalable foundation while minimizing infrastructure and operational overhead.
- The proposed solution aligns with the organization's immediate needs and provides the flexibility to evolve into more advanced architectures as business requirements grow in the future.

Key Outcomes Expected

- Reduced manual effort in email communication
- Improved accuracy and consistency
- Faster communication turnaround
- Better process standardization
- Increased operational efficiency
- Scalable solution for future business growth

7. References

Link:-

https://script.google.com/macros/s/AKfycbw2h3SZeSCleCqQOEH7EaAqiL8ZF5G4vkFvRZr6DVF56kMdm_XNEEXAN1gdJTgOx4RF4Q/exec

Source File Link:-

https://github.com/Sarvesh-co7/EMAIL_AUTOMATION_PROJECT_USING_GOOGLE_SHEETS_AND_APPSCRIPT.git

